

# **Schwartz Distribution**

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# 1 Schwartz Distribution

## I Test function and Schwartz Distribution

### i Test function

A test function on  $U$  is a functional in  $C_c^\infty(U)$ . The space of test functions on  $U$  is  $C_c^\infty(U)$ , also denoted as  $\mathcal{D}(U)$ .

### ii Set of compact subsets as directed set

The set of compact subsets of  $U$  form a directed set  $\mathbb{K}$  with order by inclusion.

### iii Topology of $C^k(K; U)$ and $C^k(U)$

For any subset  $U$  of  $\mathbb{R}^n$ , for any  $k \in \mathbb{N} \cup \{\infty\}$ , for any  $f \in C^k(U)$ , for any integer  $i$  such that  $0 \leq i \leq k$ , for any nonempty compact subset  $K$  of  $U$ , with the set  $\mathcal{P}$  of all multi-indices  $p$  such that  $|p| \leq i$ , define,

$$q_{i,K}(f) := \sup_{p \in \mathcal{P}} \left( \sup_{x \in K} |D^p f(x)| \right),$$

and define  $q_{i,\emptyset}(f) = 0$ .

Define the topology of  $C^k(K; U)$  as the topology generated by the family of seminorms  $q_{i,K}$  for any integer  $i$  such that  $0 \leq i \leq k$ , thus  $C^k(K; U)$  is a locally convex topological vector space.

Define the topology of  $C^k(U)$  as the topology generated by the family of seminorms  $q_{i,K}$  for any compact subset  $K$  of  $U$ , for any integer  $i$  such that  $0 \leq i \leq k$ , thus  $C^k(U)$  is a locally convex topological vector space.

### iv Canonical LF topology

$C_c^k(U)$  is equipped with the canonical LF topology defined as the finest topology such that for any compact subset  $K$  of  $U$ , the inclusion map from  $C^k(K; U)$  to  $C_c^k(U)$  is continuous.

### v Distribution or Schwartz distribution

A distribution or Schwartz distribution on  $U$  is a continuous linear functional on  $C_c^\infty(U)$ . The space of distributions on  $U$  is the strong dual space of  $C_c^\infty(U)$ , denoted as  $\mathcal{D}'(U)$  and equipped with the strong dual topology.

### vi Support of distribution

For any  $T \in \mathcal{D}'(U)$ , the support of  $T$  is defined as

$$\text{supp}(T) = U \setminus \bigcup \{V \subseteq U \mid \rho_{VU}(T) = 0\}.$$

If  $\text{supp}(T)$  is a compact set,  $T$  is called with compact support. If  $\text{supp}(T)$  is a compact subset of a set  $V \subseteq U$ ,  $T$  is called with compact support on  $V$ .

If  $T \in \mathcal{D}'(U)$  is with compact support, for any  $f \in C^\infty(U)$ ,  $\langle T, f \rangle$  is defined as

$$\langle T, f \rangle := \langle T, f\chi \rangle,$$

where  $\chi \in \mathcal{D}(U)$  is an arbitrary function such that there exists a neighborhood  $N$  of  $\text{supp}(T)$  such that  $\chi(N \cap U) = \{1\}$  and  $f\chi \in \mathcal{D}(U)$ . It's obvious that  $\langle T, f\chi \rangle$  doesn't depend on the choice of  $\chi$ .

## II Induced distribution

### i Distribution induced by measure

For a possibly signed Radon measure  $\mu$ , the distribution  $T_\mu \in \mathcal{D}'(U)$  induced by  $\mu$  is defined by, for all  $\varphi \in \mathcal{D}(U)$ ,

$$\langle T_\mu, \varphi \rangle = \int_U \varphi \, d\mu.$$

### ii Distribution induced by $L^1_{\text{loc}}$ function

For a function  $f \in L^1_{\text{loc}}(U)$ , the distribution  $T_f \in \mathcal{D}'(U)$  induced by  $f$  is defined by, for all  $\varphi \in \mathcal{D}(U)$ ,

$$\langle T_f, \varphi \rangle = \int_U f(x)\varphi(x) \, dx.$$

## III Extension and restriction

### i Extension and restriction

Let  $V \subseteq U \subseteq \mathbb{R}^n$ . For every function  $f \in \mathcal{D}(V)$ , the trivial extension of  $f$  to  $U$  is defined as a function  $E_{VU}(f) \in \mathcal{D}(U)$  such that for all  $v \in V$ ,

$$E_{VU}(f)(v) = f(v),$$

and for all  $u \in U \setminus V$ ,

$$E_{VU}(f)(u) = 0.$$

This defines  $E_{VU} : \mathcal{D}(V) \rightarrow \mathcal{D}(U)$  called trivial extension operator to  $U$  in  $V$ , which is a continuous injective linear map.

The transpose of  $E_{VU}$  is  $\rho_{VU} : \mathcal{D}'(U) \rightarrow \mathcal{D}'(V)$  called restriction operator to  $V$  in  $U$ , and for all distribution  $T \in \mathcal{D}'(U)$ ,  $\rho_{VU}(T)$  is called restriction of  $T$  to  $V$ . A distribution in the image of  $\rho_{VU}$  is called extendable to  $U$ . A distribution in the image of  $\rho_{V\mathbb{R}^n}$  is called extendable.

### ii Extension theorem

The image of  $E_{VU}$  is a vector space. However, for all  $V \subset U$ ,  $E_{VU}$  is not a topological embedding and that the subspace topology of the image of  $E_{VU}$  is strictly finer than the topology on the image of  $E_{VU}$  that makes  $E_{VU}$  a homeomorphism.

### iii Vanish

$T \in \mathcal{D}'(U)$  is said to vanish in a subset  $V$  of  $U$  if for all  $f \in \mathcal{D}(U)$ ,  $\langle T, f \rangle = 0$ .

### iv Gluing theorem

Let  $(U_i)_{i \in I}$  be an indexed family of open subsets of  $\mathbb{R}^n$ . For each  $i \in I$ , let  $T_i \in \mathcal{D}'(U_i)$  and suppose that for all  $i, j \in I$ , the restriction of  $T_i$  to  $U_i \cap U_j$  is equal to the restriction of  $T_j$  to  $U_i \cap U_j$ . Then there exists unique  $T \in \mathcal{D}'\left(\bigcup_{i \in I} U_i\right)$  such that for all  $i \in I$ , the restriction of  $T$  to  $U_i$  is equal to  $T_i$ .

## IV Operation

### i Multiplication by smooth function

For any open subset  $U \subseteq \mathbb{R}^n$ , for any  $T \in \mathcal{D}'(U)$ , for any  $f \in C^\infty(U)$ , the distribution  $fT$  is defined by, for any  $g \in \mathcal{D}(U)$ ,

$$\langle fT, g \rangle = \langle T, fg \rangle.$$

### ii Scaling operator

For any open subset  $U \subseteq \mathbb{R}^n$ , for any  $T \in \mathcal{D}'(U)$ ,  $T$  scaled by  $a \in \mathbb{R}$ ,  $S_a T$  or  $T(a \cdot)$ , is defined by, for any  $f \in \mathcal{D}(U)$ ,

$$\langle S_a T, f \rangle = \frac{1}{|a|^n} \langle T, f\left(\frac{\cdot}{a}\right) \rangle.$$

### iii Translation operator

For any open subset  $U \subseteq \mathbb{R}^n$ , for any  $T \in \mathcal{D}'(U)$ ,  $T$  translated by  $a \in \mathbb{R}^n$ ,  $\tau_a T$  or  $T(\cdot - a)$ , as a member of  $\mathcal{D}'(\{u + a \mid u \in U\})$ , is defined by, for any  $f \in \mathcal{D}(\{u + a \mid u \in U\})$ ,

$$\langle \tau_a T, f \rangle = \langle T, \tau_{-a} f \rangle.$$

### iv Derivative of distribution

For any open subset  $U \subseteq \mathbb{R}^n$ , for any  $T \in \mathcal{D}'(U)$ , for any  $\varphi \in \mathcal{D}(U)$ , for any multi-index  $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n)$ , the distributional derivative  $D^\alpha T$  of  $T$  is defined by,

$$\langle D^\alpha T, \varphi \rangle = (-1)^{|\alpha|} \langle T, D^\alpha \varphi \rangle.$$

### v Convolution of functionals

For any functionals  $f, g$  defined on  $\mathbb{R}^n$ , the convolution of  $f$  and  $g$ , denoted as  $f * g$ , is a functional on  $\mathbb{R}^n$  defined as

$$f * g(x) = \int_{\mathbb{R}^n} f(y)g(x - y) dy.$$

If  $\text{supp}(f) \cup \text{supp}(g) \subseteq U$ , then

$$f * g(x) = \int_U f(y)g(x - y) dy.$$

Commutativity:

$$f * g = g * f.$$

Associativity:

$$(f * g) * h = f * (g * h).$$

Translation:

$$\tau_a(f * g) = \tau_a f * g = f * \tau_a g.$$

Differentiation:

$$D^\alpha(f * g) = D^\alpha f * g = f * D^\alpha g.$$

## vi Convolution of functional and distribution

For any function  $f \in C^\infty(\mathbb{R}^n)$ , for any  $T \in \mathcal{D}'(\mathbb{R}^n)$ , with the constraint that either  $f$  or  $T$  is with compact support, the convolution of  $f$  and  $T$ , which is an element of  $C^\infty(\mathbb{R}^n)$ , denoted as  $f * T$ , is defined as

$$f * T := x \mapsto \langle T, f(x - \cdot) \rangle.$$

Translation:

$$\tau_a(f * T) = \tau_a f * T = f * \tau_a T.$$

Differentiation:

$$D^\alpha(f * T) = D^\alpha f * T = f * D^\alpha T.$$

## vii Convolution of distributions

For any open subset  $U \subseteq \mathbb{R}^n$ , for any distributions  $S, T \in \mathcal{D}'(\mathbb{R}^n)$  with the constraint that at least one of them is with compact support, for any  $f \in \mathcal{D}(\mathbb{R}^n)$ , the convolution of  $S$  and  $T$ , which is an element of  $\mathcal{D}'(U)$ , denoted as  $S * T$ , is defined as

$$\langle S * T, f \rangle := \langle S, x \mapsto \langle T, \tau_{-x} f \rangle \rangle.$$

Commutativity:

$$S * T = T * S.$$

Associativity:

$$(R * S) * T = R * (S * T).$$

Translation:

$$\tau_a(S * T) = \tau_a S * T = S * \tau_a T.$$

Differentiation:

$$D^\alpha(S * T) = D^\alpha S * T = S * D^\alpha T.$$

## viii Cross-correlation

For functionals, functional and distribution, or distributions,  $f, g$ ,

$$f \star g := (\hat{R}f)^* * g.$$

Conjugate symmetry:

$$f \star g = (\hat{R}(g \star f))^*.$$

Translation:

$$\tau_a(f * g) = \tau_{-a}f * g = f * \tau_a g.$$

Differentiation:

$$D^\alpha(f * g) = D^\alpha f * g = f * D^\alpha g.$$

## ix Fubini theorem

TODO

## V Constant one distribution

For all  $\varphi \in \mathcal{D}(U)$ ,

$$\langle 1, \varphi \rangle = \int_U \varphi(x) dx.$$

## VI Dirac delta function or unit impulse

### i Fake function

The Dirac delta function  $\delta(x)$  is often denoted as

$$\delta(x) \begin{cases} \infty, & x = 0, \\ 0, & x \neq 0. \end{cases}$$

With the derivative of unit step function  $H'(x)$  defined as  $\delta(x)$ .

### ii Stieltjes integral

For any function  $f$  defined on  $U \subseteq \mathbb{R}$ , for any indexed family with  $(A_i)_{i \in I} \subseteq U$ ,

$$\int_U f(x) \sum_{i \in I} \delta(x - A_i) dx = \int_U f(x) d \sum_{i \in I} H(x - A_i) = \sum_{i \in I} f(A_i).$$

### iii Dirac measure

Dirac (delta) measure at  $a$  is a Radon measure defined, for any set  $A$ , as

$$\delta_a(A) := \begin{cases} 1, & a \in A \\ 0, & a \notin A \end{cases}.$$

For any function  $f$  defined on  $U \subseteq \mathbb{R}^n$ , for any indexed family with  $(A_i)_{i \in I} \subseteq U$ ,

$$\int_U f(x) \sum_{i \in I} \delta_{A_i}(x) dx = \int_U f(x) d \sum_{i \in I} \delta_{A_i}(x) = \sum_{i \in I} f(A_i).$$

#### iv Dirac distribution

Dirac (delta) distribution is the distribution induced by Dirac measure, that is, for any set  $U \subseteq \mathbb{R}^n$ , for any  $a \in U$ , for any  $\varphi \in \mathcal{D}(U)$ ,

$$\langle \delta_a, \varphi \rangle = \varphi(a).$$

#### v Convolution

For all  $a \in U \subseteq \mathbb{R}^n$ , for all  $f \in \mathcal{D}(U)$ , for all  $T \in \mathcal{D}'(U)$ ,

$$f * \delta_0 = f,$$

$$T * \delta_0 = T,$$

$$f * \delta_a = \tau_a f,$$

$$T * \delta_a = \tau_a T.$$

#### vi Sinc function as nascent delta distribution

For any complex-valued or real-valued  $\varphi \in L^1(\mathbb{R})$ :

$$\lim_{a \rightarrow 0} \int_{-\infty}^{\infty} \left( \frac{\sin\left(\frac{\pi x}{a}\right)}{\pi x} \right) \varphi(y - x) dx = \varphi(y).$$

*Proof.* TODO

□

### VII Heaviside (step) distribution or unit step distribution

Heaviside (step) distribution or unit step distribution at  $a$ , denoted as  $H_a$  or  $u_a$ , is defined, for all  $U \subseteq \mathbb{R}$ , for all  $\varphi \in \mathcal{D}(U)$ , as

$$\langle H_a, \varphi \rangle = \int_{U \cap \mathbb{R}_{\geq a}} \varphi(x) dx.$$

The derivative of it is

$$H'_a = \delta_a.$$